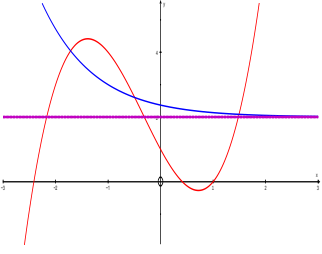


2013 SAJC H1 Math Preliminary Examination

Qn	Topic Set	Answers
1	Equations	$x = 65.9$ & $y = 54.1$
2	Inequalities	(i) $k > 7$ (ii) $x < 1$
3	Graphing	 (ii) 3.98 units^2
4	Applications of Differentiation	(ii) $x = \frac{2}{\sqrt{3}}$ maximum $V = 3\sqrt{\frac{32}{27}} \text{ cm}^3$.
5	Graphing and Applications of Differentiation	(ii) $\frac{dy}{dx} = \frac{1}{2}$ (iii) Area of triangle OPQ $= \frac{1}{2} \times 0.25 \times 3.25$ $= 0.40625 \text{ units}^2$ or $\frac{13}{32} \text{ units}^2$
6	Probability	(iia) $P(A \cap B) = 0.6p$ (iib) $0.7 + 0.4p$ $p \leq 0.75$
7	Sampling Techniques	As only the students who visited the gym get a chance of being selected, the sampling method is not random.
8	Probability	(ii) $p = 0.3$ (iii) 0.79
9	Binomial Distribution	(i) 0.00153 (iia) 0.954 (iib) 0.667
10	Correlation and Regression	(i) $k = 3.71$  (ii) (iii) $r = 0.976$. (iv) $v = 93.5 \text{ km / h}$ (v) For a), the estimate is a reliable one as the value of r is close to 1 and $v = 55$ is within the data range. For b), the estimate is not reliable as $t = 4$ is outside the data

		range.
11	Hypothesis Testing	(i) Unbiased estimate of population mean = 44.11kg Unbiased estimate of population variance 9.1525 kg (ii) $P(\bar{M} > 44.5) = 0.181$ (iii) $p\text{-value} = 0.0113$ (iv) 44.3 kg
12	Normal Distribution	(i) 0.0528 (ii) 0.984 (iii) 0.438 kg (iv) $\mu = 46$ g (nearest whole number)